

LHS Series

- The lower temperature range of the category temperature range has been expanded.
- For solar power generation
- Endurance with ripple current : 5,000 hours at 105°C
- Rated voltage range : 450 to 500V
- For inverter control, switching power supplies
- Non solvent resistant type
- RoHS2 Compliant

LHS
↑ Downsizing
LXS
↑ Longer life
KMS

500V
Lineup!

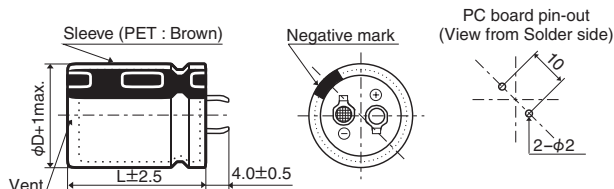


SPECIFICATIONS

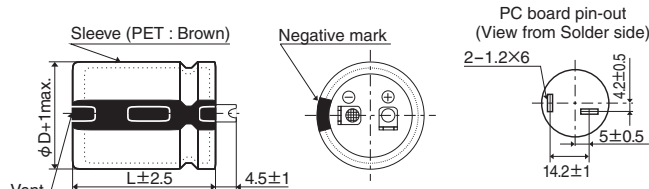
Items	Characteristics	
Category Temperature Range	-40 to +105°C	
Rated Voltage Range	450 to 500V _{dc}	
Capacitance Tolerance	±20% (M)	
Leakage Current	$I \leq 3\sqrt{CV}$ Where, I : Max. leakage current (μA), C : Nominal capacitance (μF), V : Rated voltage (V)	
Dissipation Factor (tan δ)	Rated voltage (V _{dc})	450 to 500V
	tan δ (Max.)	0.20
Low Temperature Characteristics (Max. Impedance Ratio)	Rated voltage (V _{dc})	450 to 500V
	Z(-25°C)/Z(+20°C)	8
Endurance	The following specifications shall be satisfied when the capacitors are restored to 20°C after subjected to DC voltage with the rated ripple current is applied (the peak voltage shall not exceed the rated voltage) for 5,000 hours at 105°C.	
	Capacitance change	≤ ±20% of the initial value
	D.F. (tan δ)	≤ 200% of the initial specified value (475, 500V _{dc} : ≤ 250%)
	Leakage current	≤ The initial specified value
Shelf Life	The following specifications shall be satisfied when the capacitors are restored to 20°C after exposing them for 1,000 hours at 105°C without voltage applied. Before the measurement, the capacitor shall be preconditioned by applying voltage according to Item 4.1 of JIS C 5101-4.	
	Capacitance change	≤ ±15% of the initial value
	D.F. (tan δ)	≤ 150% of the initial specified value
	Leakage current	≤ The initial specified value

DIMENSIONS [mm]

• Terminal Code : VS (φ22 to φ35) : Standard

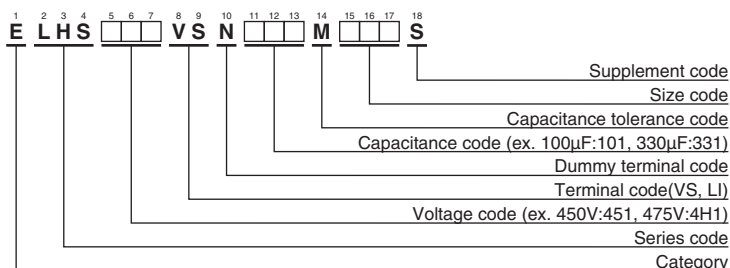


• Terminal Code : LI (φ30, φ35)



The standard design has no plastic disc.

PART NUMBERING SYSTEM



Please refer to "Product code guide (snap-in type)"



◆STANDARD RATINGS

WV (V _{dc})	Cap (μF)	Case size φD×L(mm)	tan δ	Rated ripple current (Arms/ 105°C, 120Hz)	Part No.	WV (V _{dc})	Cap (μF)	Case size φD×L(mm)	tan δ	Rated ripple current (Arms/ 105°C, 120Hz)	Part No.
450	100	22 × 25	0.20	0.71	ELHS451VSN101MP25S	475	150	30 × 25	0.20	1.01	ELHS4H1VSN151MR25S
	120	22 × 30	0.20	0.81	ELHS451VSN121MP30S		180	30 × 30	0.20	1.11	ELHS4H1VSN181MR30S
	150	22 × 35	0.20	0.93	ELHS451VSN151MP35S		180	35 × 25	0.20	1.08	ELHS4H1VSN181MA25S
	150	25.4 × 25	0.20	0.93	ELHS451VSN151MQ25S		220	30 × 35	0.20	1.26	ELHS4H1VSN221MR35S
	180	22 × 40	0.20	1.04	ELHS451VSN181MP40S		270	30 × 40	0.20	1.44	ELHS4H1VSN271MR40S
	180	25.4 × 30	0.20	1.05	ELHS451VSN181MQ30S		270	35 × 30	0.20	1.35	ELHS4H1VSN271MA30S
	220	22 × 45	0.20	1.17	ELHS451VSN221MP45S		330	30 × 45	0.20	1.63	ELHS4H1VSN331MR45S
	220	25.4 × 35	0.20	1.21	ELHS451VSN221MQ35S		330	35 × 35	0.20	1.51	ELHS4H1VSN331MA35S
	220	30 × 25	0.20	1.15	ELHS451VSN221MR25S		390	30 × 50	0.20	1.80	ELHS4H1VSN391MR50S
	270	22 × 50	0.20	1.33	ELHS451VSN271MP50S		390	35 × 40	0.20	1.70	ELHS4H1VSN391MA40S
	270	25.4 × 40	0.20	1.36	ELHS451VSN271MQ40S		470	30 × 60	0.20	2.05	ELHS4H1VSN471MR60S
	270	30 × 30	0.20	1.29	ELHS451VSN271MR30S		470	35 × 45	0.20	1.91	ELHS4H1VSN471MA45S
	270	35 × 25	0.20	1.25	ELHS451VSN271MA25S		470	35 × 50	0.20	1.95	ELHS4H1VSN471MA50S
	330	22 × 60	0.20	1.54	ELHS451VSN331MP60S	500	560	35 × 60	0.20	2.21	ELHS4H1VSN561MA60S
	330	25.4 × 45	0.20	1.54	ELHS451VSN331MQ45S		120	30 × 25	0.20	0.90	ELHS501VSN121MR25S
	330	25.4 × 50	0.20	1.56	ELHS451VSN331MQ50S		150	30 × 30	0.20	1.02	ELHS501VSN151MR30S
	330	30 × 35	0.20	1.46	ELHS451VSN331MR35S		150	35 × 25	0.20	0.99	ELHS501VSN151MA25S
	330	35 × 30	0.20	1.41	ELHS451VSN331MA30S		180	30 × 35	0.20	1.14	ELHS501VSN181MR35S
	390	25.4 × 60	0.20	1.74	ELHS451VSN391MQ60S		220	30 × 40	0.20	1.30	ELHS501VSN221MR40S
	390	30 × 40	0.20	1.63	ELHS451VSN391MR40S		220	35 × 30	0.20	1.22	ELHS501VSN221MA30S
	470	30 × 45	0.20	1.84	ELHS451VSN471MR45S		270	30 × 45	0.20	1.47	ELHS501VSN271MR45S
	470	30 × 50	0.20	1.87	ELHS451VSN471MR50S		270	35 × 35	0.20	1.37	ELHS501VSN271MA35S
	470	35 × 35	0.20	1.71	ELHS451VSN471MA35S		330	30 × 50	0.20	1.66	ELHS501VSN331MR50S
	560	35 × 40	0.20	1.95	ELHS451VSN561MA40S		330	35 × 40	0.20	1.57	ELHS501VSN331MA40S
	560	35 × 45	0.20	1.99	ELHS451VSN561MA45S		390	30 × 60	0.20	1.87	ELHS501VSN391MR60S
	680	30 × 60	0.20	2.33	ELHS451VSN681MR60S		390	35 × 45	0.20	1.74	ELHS501VSN391MA45S
	680	35 × 50	0.20	2.22	ELHS451VSN681MA50S		470	35 × 50	0.20	1.95	ELHS501VSN471MA50S
	820	35 × 60	0.20	2.52	ELHS451VSN821MA60S		560	35 × 60	0.20	2.21	ELHS501VSN561MA60S

◆RATED RIPPLE CURRENT MULTIPLIERS

● Frequency Multipliers

Frequency(Hz)	50	120	300	1k	10k	50k
450V _{dc}	0.77	1.00	1.16	1.30	1.41	1.43
475, 500V _{dc}	0.70	1.00	1.16	1.30	1.41	1.43

The deterioration of aluminum electrolytic capacitors accelerates their life due to the internal heating produced by ripple current. For details, refer to Section "5-3 Ripple Current Effect on Lifetime" in the catalog, Technical Note.



- Always read "Notes on Use" before using the product in order to enable you to use the product correctly and prevent any faults and accidents from occurring.
- Request the Product Specification on the product of NIPPON CHEMI-CON CORPORATION to refer to it as well as this brochure prior to the order of the products. Some specific notes on use of the ordered product may be described in the specifications.
- The products listed in this catalog are designed and manufactured for general electronics equipment use and are not intended for use in applications that can adversely affect human life; where the malfunction of equipment may cause damage to life or property. In addition, our products are not intended to be used in specific applications that may cause a major social impact. Please consult with us in advance of usage of our products in the following listed applications. ① Aerospace equipment ② Power generation equipment such as thermal power, nuclear power etc. ③ Medical equipment ④ Transport equipment (automobiles, trains, ships, etc.) ⑤ Transportation control equipment ⑥ Disaster prevention / crime prevention equipment ⑦ Highly publicized information processing equipment ⑧ Submarine equipment ⑨ Other applications that are not considered general-purpose applications.
- The circuits described as examples in this catalog and the "delivery specifications" are featured in order to show the operations and usage of our products, however, this fact does not guarantee that the circuits are available to function in your equipment systems. We are not in any case responsible for any failures or damage caused by the use of information contained herein. You should examine our products, of which the characteristics are described in the "delivery specifications" and other documents, and determine whether or not our products suit your requirements according to the specifications of your equipment systems. Therefore, you bear final responsibility regarding the use of our products.
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- We strongly recommend our customers to purchase Nippon Chemi-Con products only through our official sales channels. We assume no responsibility for any defects or damages caused by using products purchased from outside our official sales channel or of counterfeit goods. In addition, we will ask the customer to pay the investigation cost for products purchased outside our official sales channel.
- We reserve the right to discontinue production and delivery of products. We do not guarantee that all the products included in this catalog will be available in the future.
The aforementioned does not apply in the case of individual agreements deviating from the foregoing for customer-specific products
- We continually strive to improve the quality and reliability of our products, but in any case that our product does not meet our published specifications, please stop using it promptly and contact us immediately. As for compensation for non-conforming goods delivered by Chemi-Con, we will limit it only to goods found in non-compliance of our published specifications. This may be accomplished by a no cost replacement of non-conforming individual products, a credit of the piece price paid per each individual non-conforming product, or in other ways deemed necessary.
In addition, we have an established system with enhanced traceability, therefore we will limit the applicable lot items for any potential compensation.

[Part Numbering System](#)

[Part Numbering System \(Appendix\)](#)

[Standardization](#)

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[Available Terminals for Snap-in and Screw Mount Type](#)